Golden field vs recomputed field — 2025-06-01 00 UTC

Field statistics (gpm)

	min	max	range	mean	std	p05	p95	mean |value|
Grams golden	-181.1	148.7	329.7	-26.2	82.4	-157.1	96.7	72.8
ERA5 − WB2 seasonal	-194.5	135.2	329.6	-32.2	84.1	-167.6	93.0	74.9
ERA5 − year-round	-70.3	291.9	362.2	63.1	72.2	-46.5	184.3	76.6

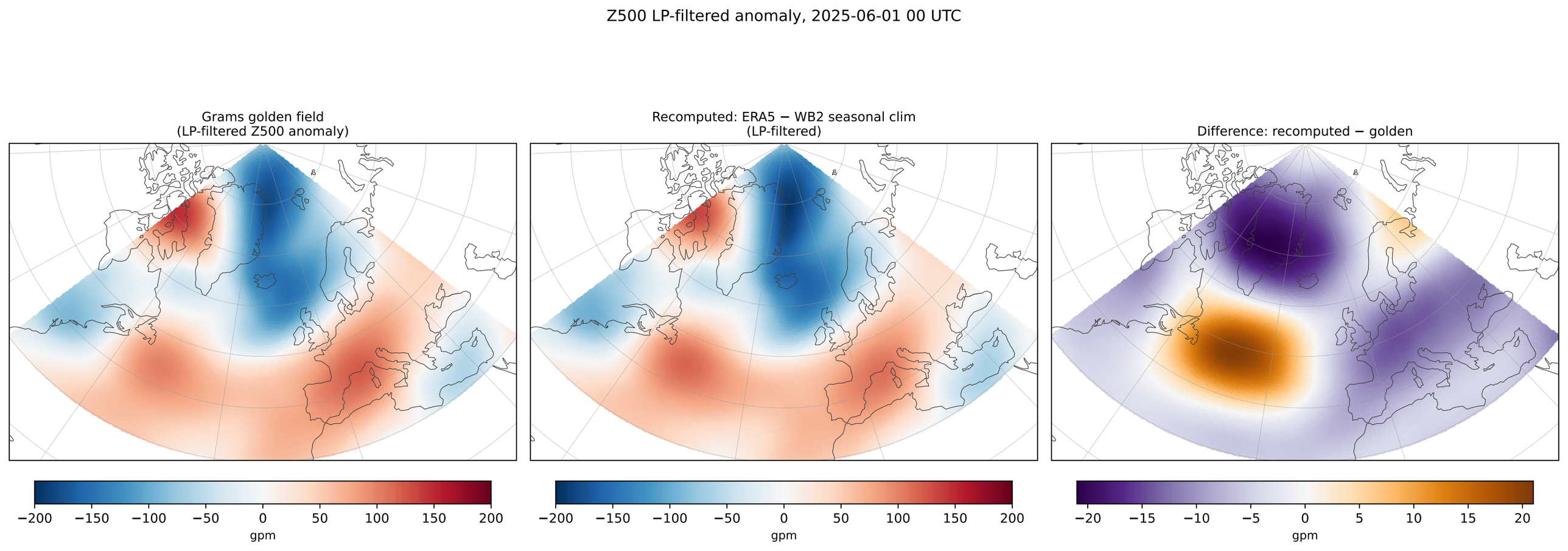
Errors vs golden field (gpm)

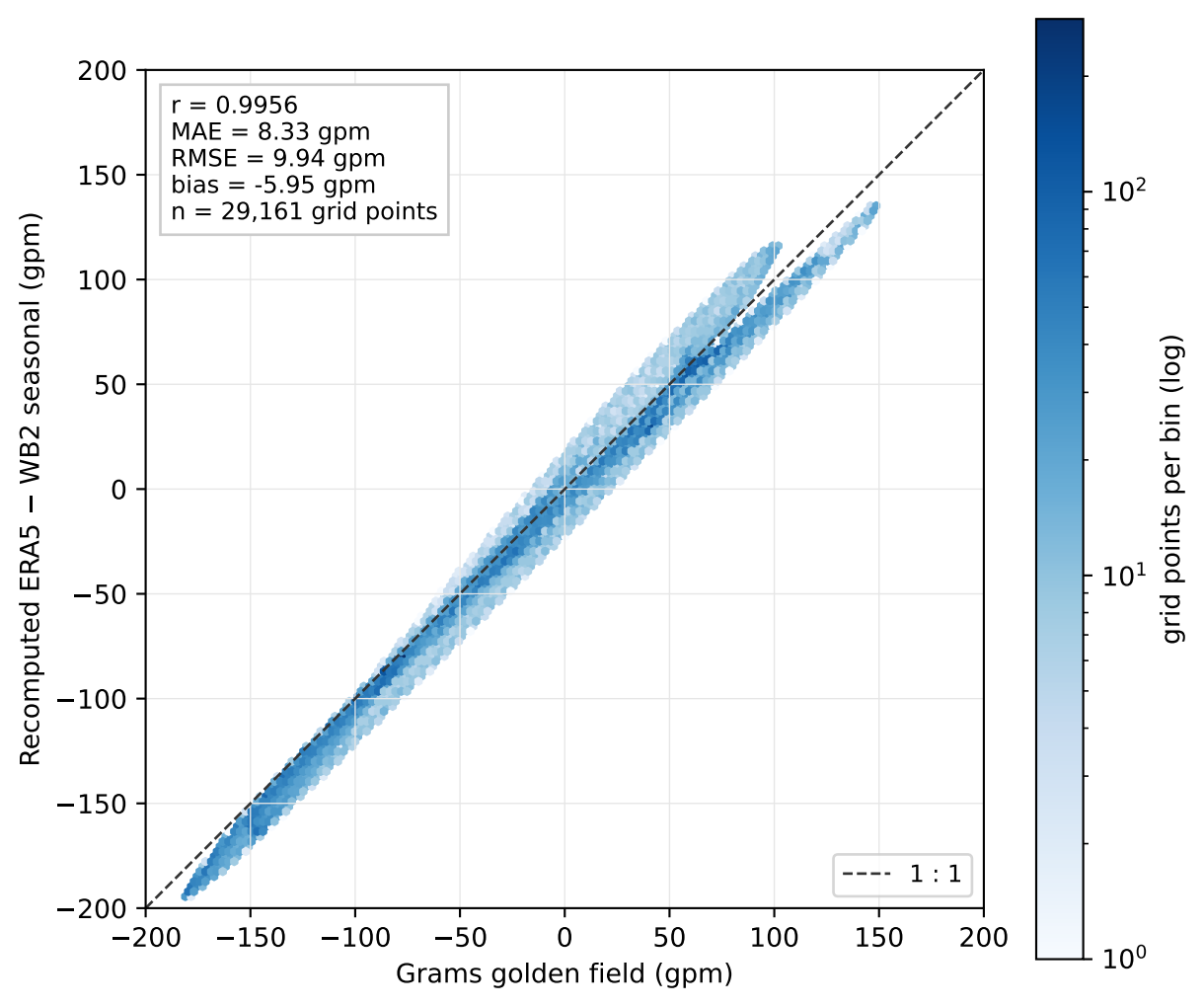
	bias	MAE	RMSE	max |diff|	MAE (cos-w)	RMSE (cos-w)	pattern r
ERA5 − WB2 seasonal	-5.951	8.335	9.943	21.094	8.003	9.450	0.996
ERA5 − year-round	89.323	89.323	97.528	145.616	71.824	81.596	0.880

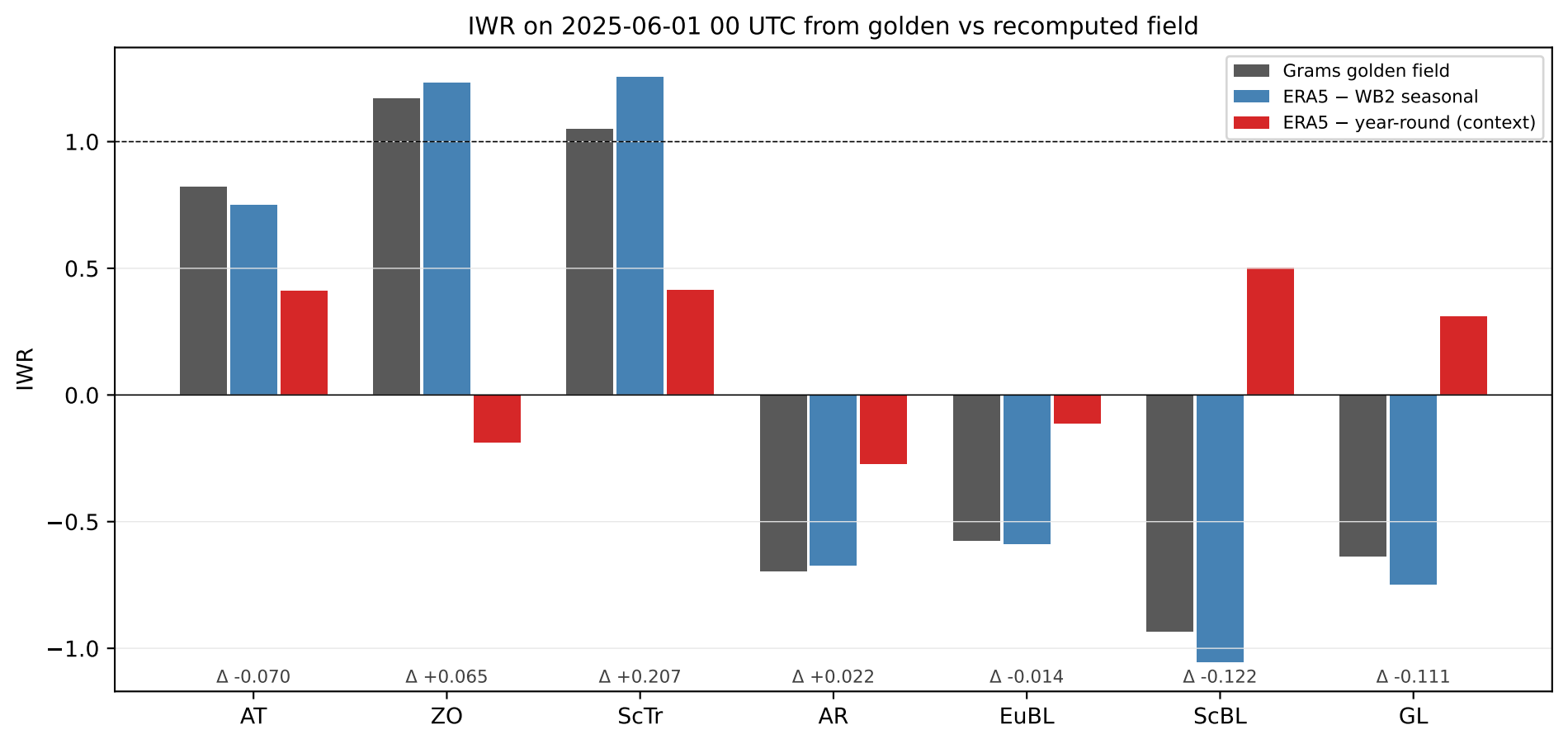
Errors relative to the golden field

	MAE / range	RMSE / range	RMSE / std
ERA5 − WB2 seasonal	2.53 %	3.02 %	12.07 %
ERA5 − year-round	27.09 %	29.58 %	118.37 %

Golden field (Z0500_20250601_00.nc, EOF domain 30–90°N, 80°W–40°E, 29,161 points): values from -181.1 to 148.7 gpm (range 329.7 gpm, std 82.4 gpm).
Recomputed from ERA5 with the WeatherBench2 seasonal climatology (1990–2019): MAE 8.33 gpm (2.53 % of range), RMSE 9.94 gpm (3.02 % of range, 12.1 % of std), bias -5.95 gpm, pattern r = 0.9956.
IWR: mean |Δ| over the 7 regimes = 0.087 (year-round mean: 0.810).


Z500 LP-filtered anomaly, 2025-06-01 00 UTC
Grams golden field
(LP-filtered Z500 anomaly)
Recomputed: ERA5 − WB2 seasonal clim
(LP-filtered)
Difference: recomputed − golden
−200 −150 −100 −50 0 50 100 150 200
gpm
−200 −150 −100 −50 0 50 100 150 200
gpm
−20 −15 −10 −5 0 5 10 15 20
gpm


r = 0.9956
MAE = 8.33 gpm
RMSE = 9.94 gpm
bias = -5.95 gpm
n = 29,161 grid points
1 : 1
Recomputed ERA5 − WB2 seasonal (gpm)
Grams golden field (gpm)
grid points per bin (log)
200
150
100
50
0
−50
−100
−150
−200
10^2
10^1
10^0


IWR on 2025-06-01 00 UTC from golden vs recomputed field
Grams golden field
ERA5 − WB2 seasonal
ERA5 − year-round (context)
IWR
1.0
0.5
0.0
−0.5
−1.0
Δ -0.070
Δ +0.065
Δ +0.207
Δ +0.022
Δ -0.014
Δ -0.122
Δ -0.111
AT
ZO
ScTr
AR
EuBL
ScBL
GL

IWR comparison

IWR per regime

	Grams published	golden field	ERA5 − WB2 seasona	ERA5 − year-round	Δ WB2 seasonal	Δ year-round
AT	+0.822	+0.822	+0.752	+0.410	-0.070	-0.412
ZO	+1.169	+1.169	+1.234	-0.189	+0.065	-1.358
ScTr	+1.049	+1.049	+1.256	+0.416	+0.207	-0.633
AR	-0.695	-0.695	-0.673	-0.271	+0.022	+0.424
EuBL	-0.574	-0.574	-0.588	-0.112	-0.014	+0.462
ScBL	-0.932	-0.932	-1.055	+0.501	-0.122	+1.434
GL	-0.637	-0.637	-0.748	+0.311	-0.111	+0.948